Sales Data Analysis Project

Overview

This project presents a complete Exploratory Data Analysis (EDA) of a retail sales dataset. The objective is to clean raw data, analyze customer purchasing behavior, and extract meaningful business insights using Python.

The workflow includes data cleaning, preprocessing, exploratory data analysis, data visualization, and business insight generation to support data-driven decision-making.

Objectives

- Clean and prepare raw sales data for analysis.
- Handle missing values, duplicate records, and inconsistent formatting.
- Explore customer behavior and sales trends.
- Analyze relationships between variables.
- Generate actionable business insights through data visualization.

Dataset Description

The dataset contains retail sales transactions with the following features:

- Order ID
- Order Date
- City
- Product Category
- Payment Method
- Customer Age
- Order Total
- Customer Rating

Tools & Libraries

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

Data Cleaning Steps

The following preprocessing steps were performed:

- Removed duplicate records.
- Filled missing numerical values using median imputation.
- Converted incorrect data types (dates and IDs).
- Standardized formatting for cities, payment methods, and product categories.
- Removed invalid values such as negative order totals and invalid customer ages.
- Detected and analyzed outliers using the IQR method.
- Created age groups for additional customer analysis.

Exploratory Data Analysis (EDA)

Correlation Analysis

A correlation heatmap was used to identify relationships between numerical variables.

Customer Age vs. Spending

Customer age was compared with order total using scatter plots and a regression line to evaluate whether age influences spending behavior.

Distribution Analysis

The distributions of the following variables were analyzed:

- Customer Age
- Order Total

Histograms and boxplots were used to understand data distribution and identify potential outliers.

Payment Behavior

Payment methods were analyzed to understand customer payment preferences and how they vary across different cities.

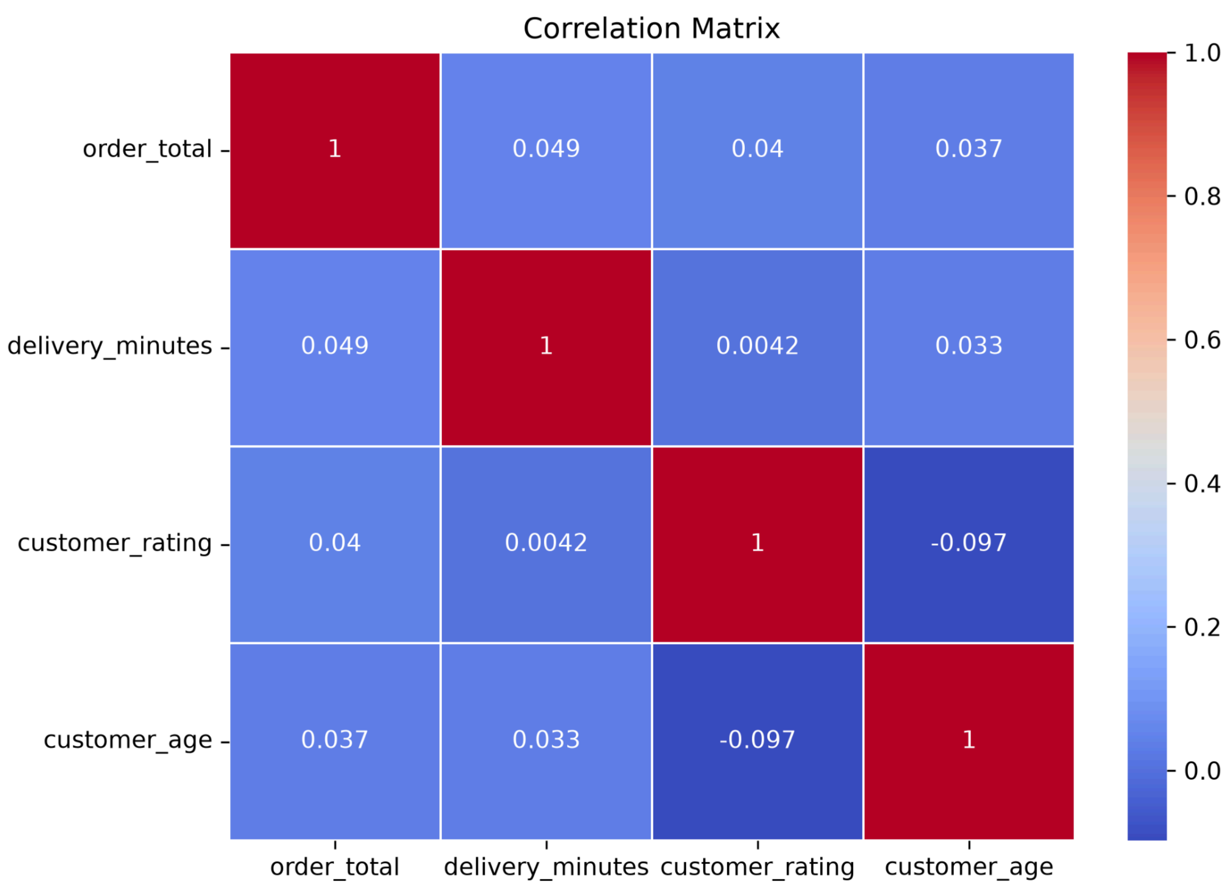
Revenue Analysis

Revenue was analyzed by:

- Product Category
- City

Results and Visualizations

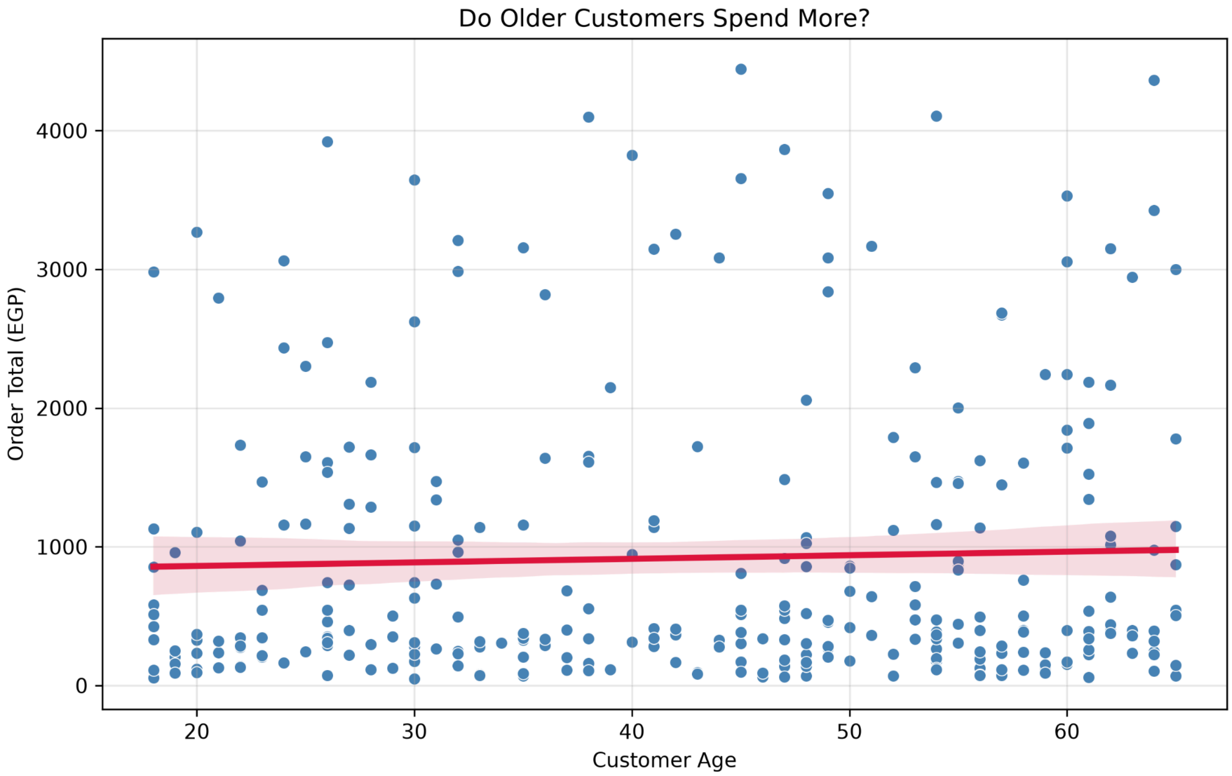
Figure 1: Correlation Matrix



Interpretation

The correlation matrix illustrates the relationships between numerical variables in the dataset. It helps identify whether variables have positive, negative, or weak correlations. In this project, customer age and order total show only a weak positive relationship.

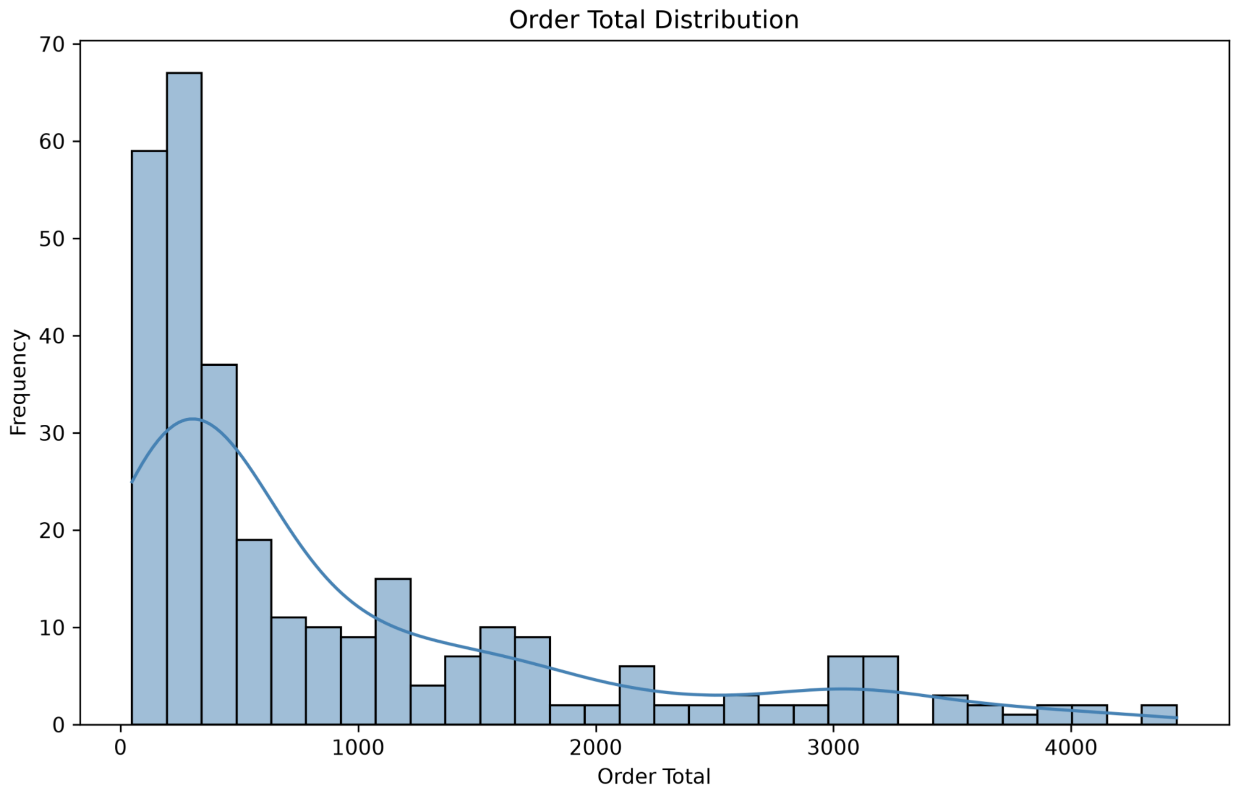
Figure 2: Customer Age vs. Order Total



Interpretation

The scatter plot and regression line suggest a weak positive relationship between customer age and spending. Although older customers tend to spend slightly more on average, the relationship is not strong enough to conclude that age is a major factor affecting purchase value.

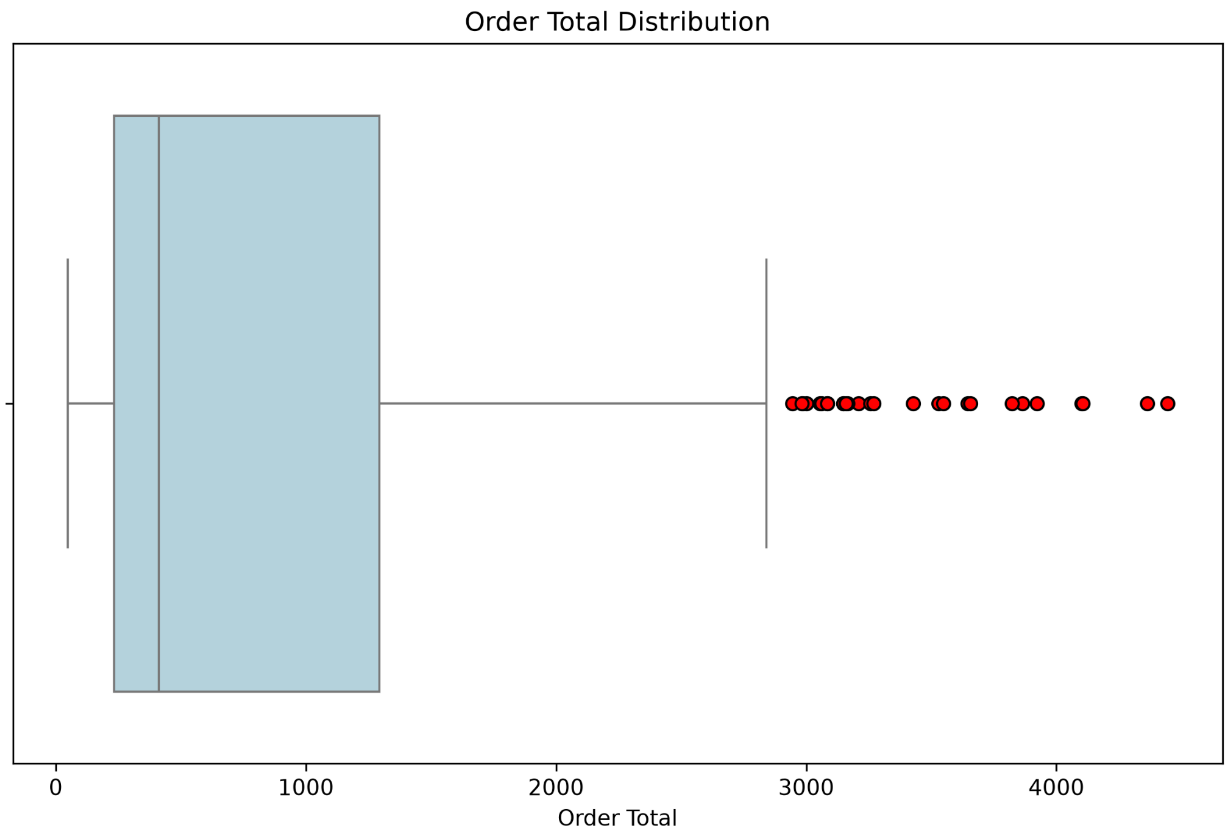
Figure 3: Order Total Distribution (Histogram)



Interpretation

The histogram shows the distribution of order totals. Most purchases are concentrated within the lower and middle price ranges, while fewer transactions occur at very high values.

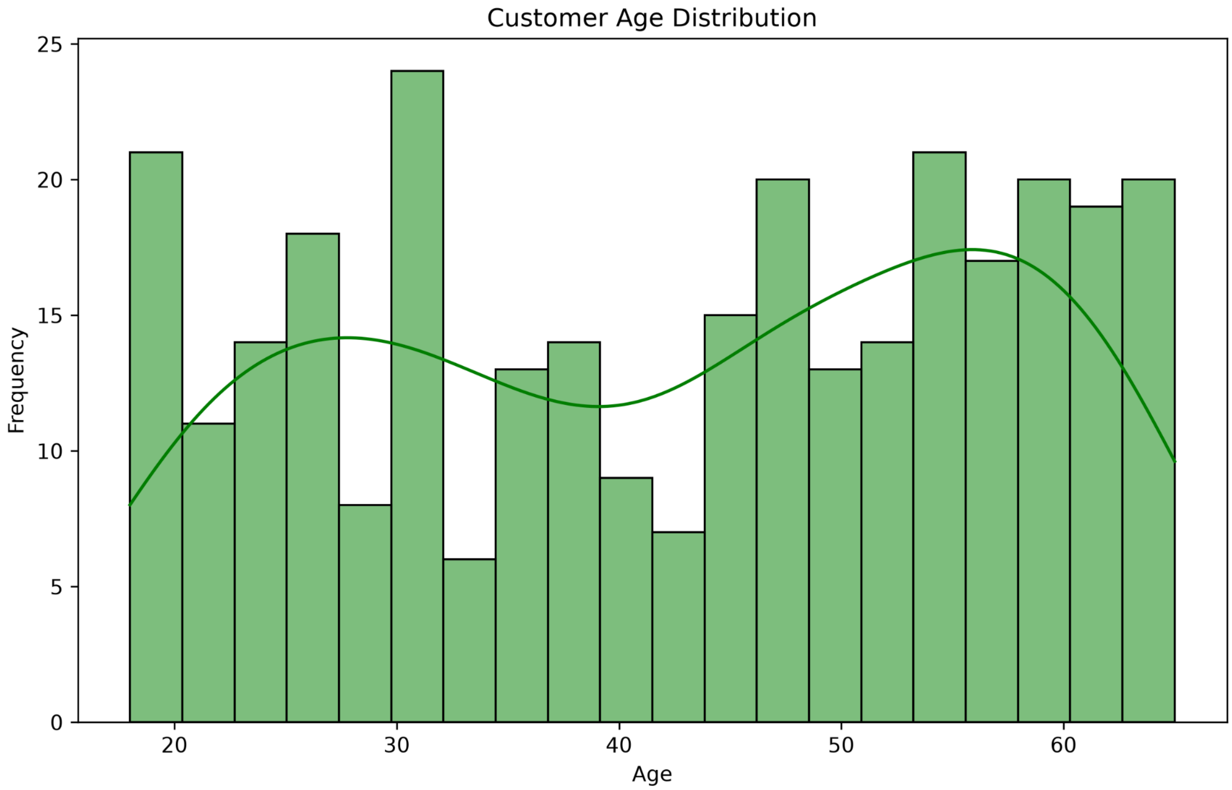
Figure 4: Order Total Boxplot



Interpretation

The boxplot highlights the spread of order totals and identifies several outliers. These high-value purchases are likely to represent premium transactions rather than data entry errors.

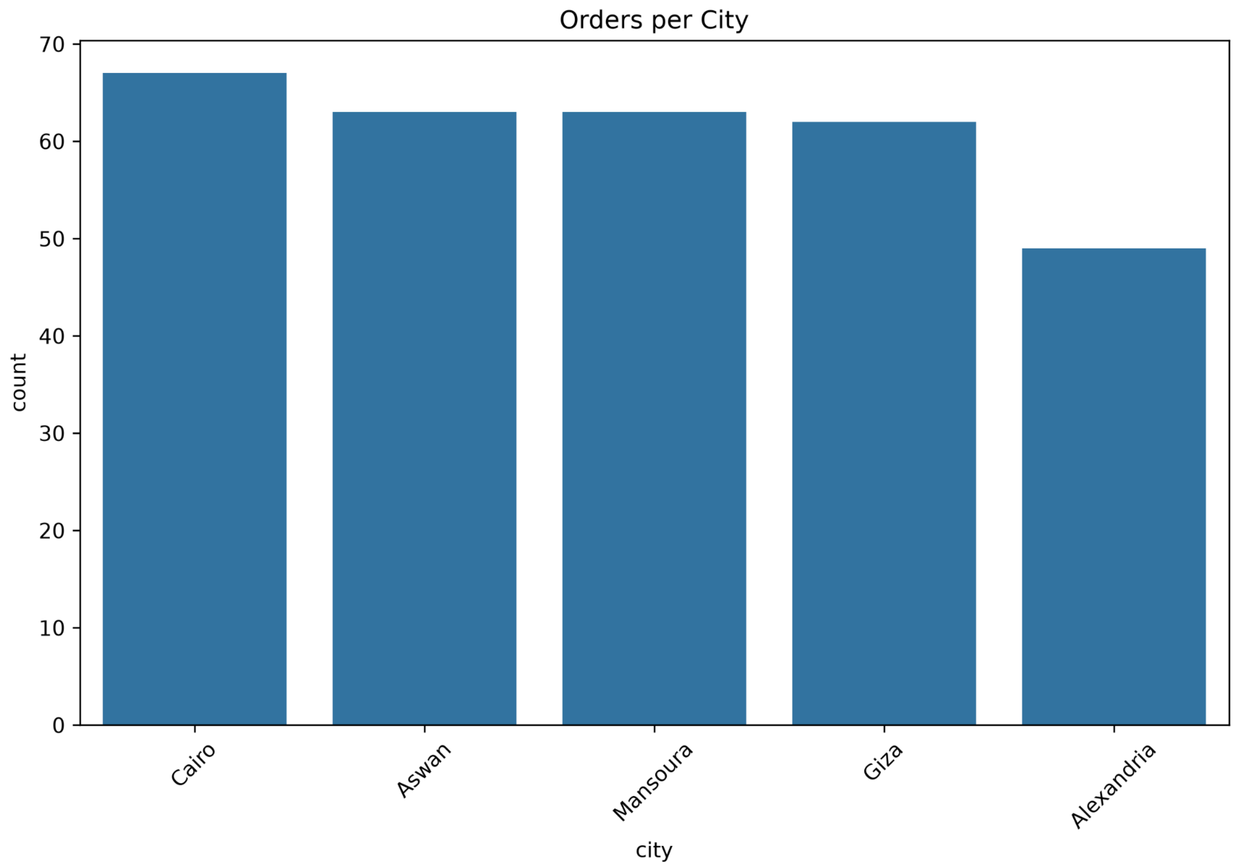
Figure 5: Customer Age Distribution



Interpretation

The age distribution indicates that most customers belong to the middle-aged and old groups, suggesting that these age groups represent the primary customer base.

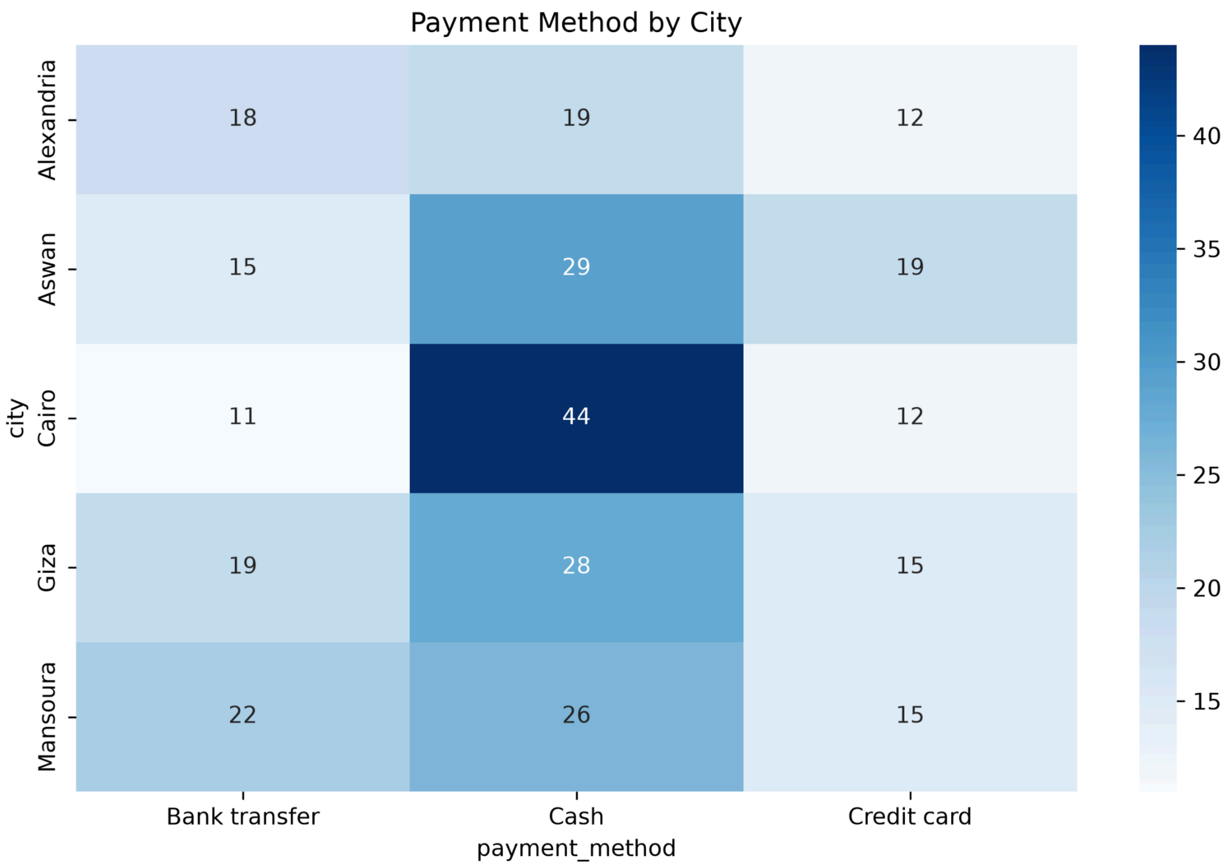
Figure 6: City Distribution



Interpretation

This chart displays the number of orders placed in each city, allowing comparison of customer activity across different locations.

Figure 7: Payment Method by City

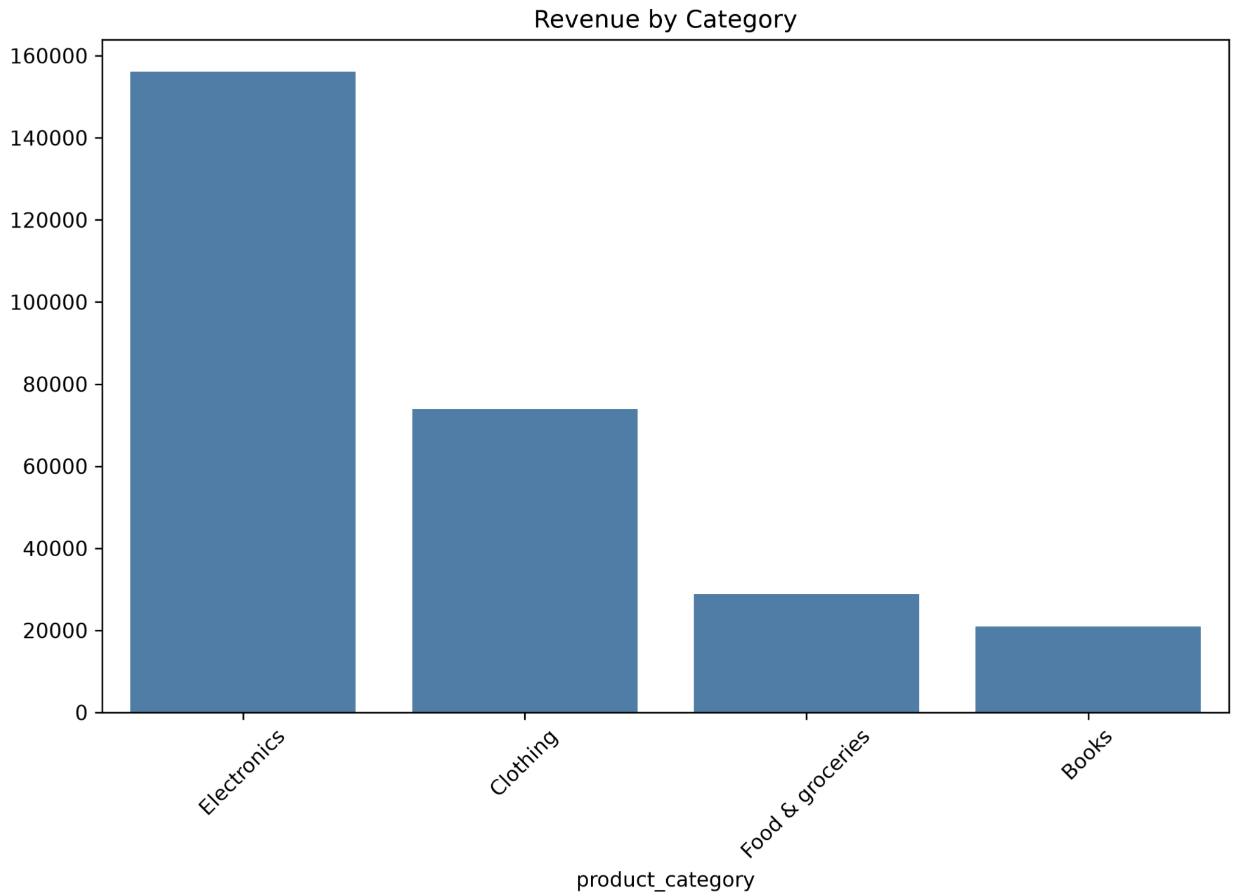


Interpretation

The heatmap compares payment methods across cities. It helps identify cairo residents' preference of paying using cash.

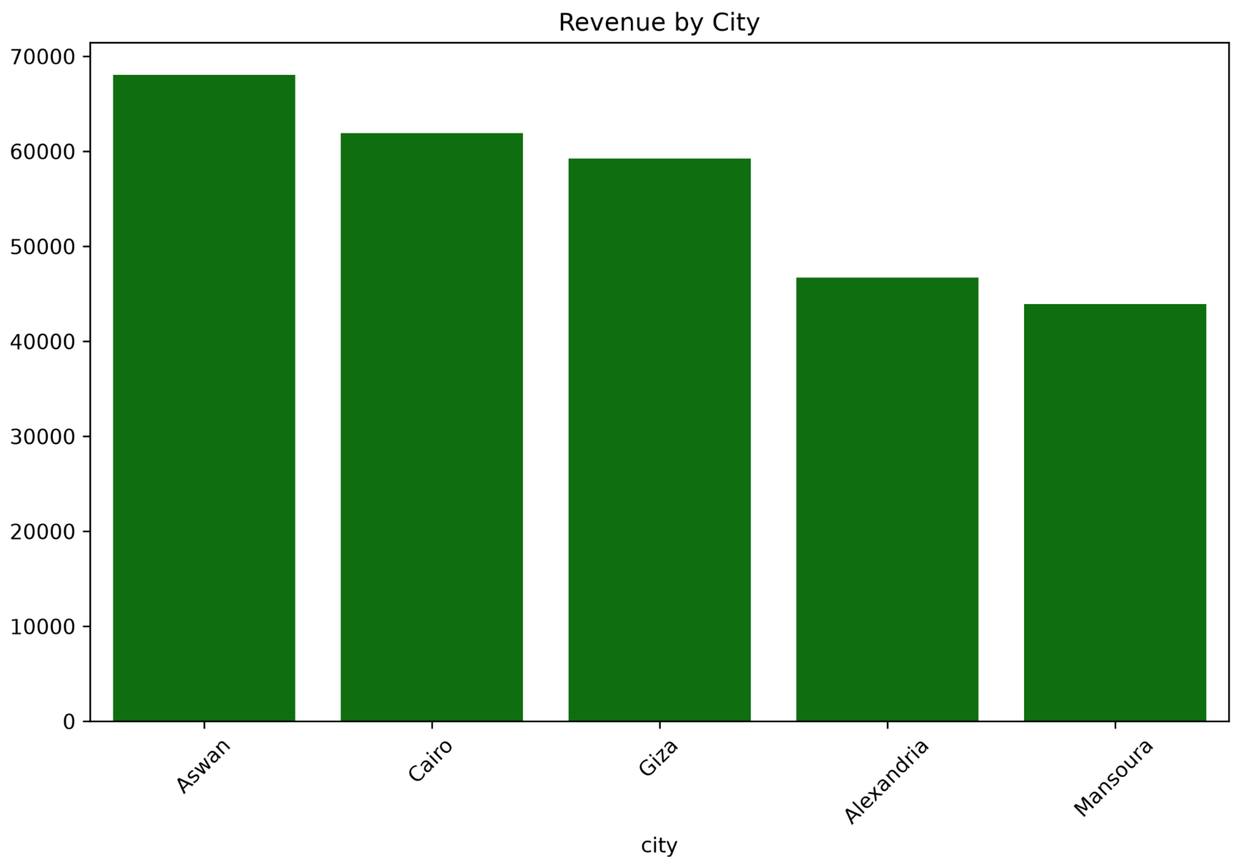
Figure 8: Revenue by Product Category

Interpretation



The chart compares total revenue generated by each product category. Electronics generate the highest revenue, likely because electronic products generally have higher prices than other product categories.

Figure 9: Revenue by City



Interpretation

This visualization compares total revenue generated by each city. It highlights the lower revenue from Alexandria and mansoura

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Key Insights

- A weak positive correlation exists between customer age and spending.
- Cash is the most commonly used payment method.
- Revenue varies significantly across product categories.
- Certain cities generate considerably higher revenue than others.
- Most customers belong to the young and middle-aged age groups.

- High-value order outliers are likely to represent premium purchases rather than data errors.

Visualizations

The project includes the following visualizations:

- Correlation Matrix
- Customer Age vs. Order Total (Scatter Plot)
- Order Total Distribution (Histogram)
- Customer Age Distribution
- Order Total Boxplot with Outliers
- Payment Method Distribution
- City Distribution
- Payment Method by City Heatmap
- Revenue by Product Category
- Revenue by City
- Payment Method Share

Business Hypotheses

Hypothesis 1

Higher cash payment rates may be influenced by customer payment preferences, limited availability of digital payment options, or local purchasing habits.

Hypothesis 2

Electronics generate the highest revenue because electronic products generally have higher selling prices than products in other categories, resulting in greater overall sales revenue.

Hypothesis 3

Mansoura have significantly lower revenue although it has average order rates ,this may be caused the residents aren’t buying high revenue products

Hypothesis 4

Alexandria has lower order and revenue this may be caused by lack of support and care for the city or not offering high demand products

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Conclusion

This project demonstrates how raw sales data can be transformed into meaningful business insights through effective data cleaning, exploratory data analysis, and visualization. The findings highlight customer behavior, payment preferences, and revenue trends, providing valuable information to support business decision-making.

Future Improvements

- Perform statistical tests to validate analytical findings.
- Develop interactive dashboards using Power BI.
- Analyze larger and more diverse datasets.
- Explore additional business questions for deeper insights.